

Beyond Spherical Symmetry: A Weakened Condition Toward Feasible Universal Feature Selection

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1. Introduction

- Feature selection [1] is crucial in high-dimensional learning and inference.
- Task-specific features can be obtained with sufficient domain knowledge.
- A more ambitious goal is to extract **universally effective** features that remain applicable across a diverse range of tasks.
- For Markov chain framework $U \leftrightarrow X \leftrightarrow Y \leftrightarrow V$, we extract features based on observed variables X, Y to infer the latent attributes U, V .

2. Preliminaries

- ϵ -neighborhood: $\mathcal{N}_Z^\epsilon(P_Z) = \{P \in \mathcal{P}^Z : \chi^2(P \| P_Z) \leq \epsilon^2\}$.
- ϵ -attribute: W is an ϵ -attribute of Z , if $P_{Z|W}(\cdot | w) \in \mathcal{N}_Z^\epsilon(P_Z)$ for any $w \in \mathcal{W}$. Its configuration $\mathcal{C}_Z^\epsilon(P_Z) = \{\mathcal{W}, \{P_W(w)\}, \{P_{Z|W}(\cdot | w)\}\}$.
- Information matrix & vector: For $z_i \in \mathcal{Z}, w_j \in \mathcal{W}$,

$$\Phi_{i,j}^{Z|W} = \frac{1}{\epsilon} \cdot \frac{P_{Z|W}(z_i | w_j) - P_Z(z_i)}{\sqrt{P_Z(z_i)}}.$$

The j -th column $\phi_{w_j}^{Z|W}$ is the information vector for w_j .

- Feature vectors: Given normalized features $h^k = (h_1, \dots, h_k)^T : \mathcal{Z} \rightarrow \mathbb{R}^k$, define $\psi_i^Z(z) = \sqrt{P_Z(z)} h_i(z), i = 1, \dots, k$.
- Canonical dependence matrix: For X, Y , let $K_{XY} = \min\{|\mathcal{X}|, |\mathcal{Y}|\}$ and

$$(\tilde{B}_{X,Y})_{j,i} = \frac{P_{X,Y}(x_i, y_j) - P_X(x_i)P_Y(y_j)}{\sqrt{P_X(x_i)}\sqrt{P_Y(y_j)}}.$$

3. Problem Setup

- Markov chain $U \xleftrightarrow{\epsilon} X \leftrightarrow Y \xleftrightarrow{\epsilon} V$, where U, V are the ϵ -attributes of X, Y .
- Data $x^N = (x_1, \dots, x_N), y^N = (y_1, \dots, y_N)$ i.i.d. drawn from $P_{X,Y}$.
- Statistics S^k and T^k are the empirical expectations of $f^k(X)$ and $g^k(Y)$.
- The collection of configurations $\mathcal{C}_X^\epsilon(P_X)$ and $\mathcal{C}_Y^\epsilon(P_Y)$ are equipped with probability measures μ_U and μ_V .
- Denote the error probability for inferring U from S^k by $p_e^{U|S}(f^k, \mathcal{C}_X^\epsilon(P_X))$, and the Chernoff error exponent

$$\bar{E}^{U|S}(f^k) = \lim_{N \rightarrow \infty} -\frac{\mathbb{E}_{\mu_U}[\log p_e^{U|S}(f^k, \mathcal{C}_X^\epsilon(P_X))]}{N}.$$

Similarly define $\bar{E}^{V|S}(f^k), \bar{E}^{U|T}(g^k)$ and $\bar{E}^{V|T}(g^k)$.

- We aim to seek features f^k and g^k that maximize the above exponents.

4. Previous Results

Recent works [2, The 1] and [3, Prop 5.10] propose that it's natural to assume μ_U and μ_V exhibit **no directional preference**, i.e. $\Phi^{X|U}$ and $\Phi^{Y|V}$ satisfy spherical symmetry [4], also called rotational invariance.

Definition 1 (Spherical symmetry) A random matrix $A \in \mathbb{R}^{n \times m}$ is spherically symmetric, if for any orthogonal matrices $Q_1 \in \mathbb{R}^{n \times n}$ and $Q_2 \in \mathbb{R}^{m \times m}$, we have $Q_1^T A Q_2 \stackrel{d}{=} A$.

Theorem 1 (Huang et al. 2024) For $k \in \{1, \dots, K_{XY} - 1\}$, select normalized features $f^k : \mathcal{X} \rightarrow \mathbb{R}^k$ and $g^k : \mathcal{Y} \rightarrow \mathbb{R}^k$. Assume under each probability measures, both $\Phi^{X|U}$ and $\Phi^{Y|V}$ are spherically symmetric. Then,

$$(\bar{E}^{U|S}(f^k), \bar{E}^{V|S}(f^k), \bar{E}^{U|T}(g^k), \bar{E}^{V|T}(g^k)) \leq (C_U \epsilon^2 k, C_V \epsilon^2 \sum_{i=1}^k \sigma_i^2, C_U \epsilon^2 \sum_{i=1}^k \sigma_i^2, C_V \epsilon^2 k) + o(\epsilon^2), \quad (1)$$

as $\epsilon \rightarrow 0$, where C_U and C_V are positive constants independent of ϵ, k or $P_{X,Y}$, and σ_i is the i -th largest singular value of $\tilde{B}_{X,Y}$. Moreover, the equality holds when the feature vectors corresponding to f_i and g_i are precisely the right and left singular vectors of $\tilde{B}_{X,Y}$ corresponding to σ_i .

5. Motivation and Observation

The exact spherical symmetry is rather restrictive and challenging to achieve:

Lemma 2 If A is spherically symmetric, then: (1) $\mathbb{E}[A] = 0$; (2) For any $i, j, k, l, A_{i,j} \stackrel{d}{=} A_{k,l}$; (3) For $(i, j) \neq (k, l)$, $\text{Cov}(A_{i,j}, A_{k,l}) = 0$.

We have the following observation. Assume W is an ϵ -attribute of Z , then

$$\log \frac{P_{Z|W}(z | w)}{P_Z(z)} = \epsilon \frac{\phi_w(z)}{\sqrt{P_Z(z)}} - \frac{\epsilon^2}{2} \left(\frac{\phi_w(z)}{\sqrt{P_Z(z)}} \right)^2 + o(\epsilon^2),$$

where ϕ_w is the information vector for w . For any smooth one-parameter family $\{w(\theta)\}$ passing through w at $\theta = 0$, the score function satisfies

$$\left. \frac{\partial}{\partial \theta} \log P(z | w(\theta)) \right|_{\theta=0} = \epsilon \frac{\phi'_w(z)}{\sqrt{P_Z(z)}} + O(\epsilon^2),$$

where ϕ'_w denotes the derivative in the direction of θ . Then the Fisher information becomes $I(0) = \epsilon^2 \|\phi'_w\|^2 + O(\epsilon^3)$. According to the **Cramér–Rao lower bound**, as $\epsilon \rightarrow 0$, the term $\|\phi'_w\|^2$ dominates the mean-square error of estimation, indicating that in the local regime, imposing spherical symmetry can be reduced to a **second-moment symmetry condition**.

6. Main Contributions

We propose weak spherical symmetry based on second-moment deviation.

Definition 2 (Weak spherical symmetry) For random matrices $X, Y \in \mathbb{R}^{n \times m}$, define a second-moment distance as

$$D(X, Y) = \sup_{u \in \mathbb{R}^n, v \in \mathbb{R}^m, \|u\|=\|v\|=1} \left| \mathbb{E}[(u^T X v)^2] - \mathbb{E}[(u^T Y v)^2] \right|.$$

Given $\delta \geq 0$, a random matrix $A \in \mathbb{R}^{n \times m}$ is δ -spherically symmetric, if for any orthogonal matrices $Q_1 \in \mathbb{R}^{n \times n}, Q_2 \in \mathbb{R}^{m \times m}$, we have $D(Q_1^T A Q_2, A) \leq \delta$.

Remark 1 Spherical symmetry indicates δ -spherical symmetry with $\delta = 0$.

Example 1 Given $0 < |\sigma^2 - 1| \leq \delta$. A random matrix $A \in \mathbb{R}^{2 \times 2}$ has independent entries $A_{1,1} \sim N(0, \sigma^2)$ and $A_{1,2}, A_{2,1}, A_{2,2} \sim N(0, 1)$. Then, A is δ -spherically symmetric, but not exactly spherically symmetric.

The weakened condition allows bounded ($\leq \delta$) directional preference to exist. Under this condition, we demonstrate the following theorem.

Theorem 3 (Tang, Han, 2026) For $k \in \{1, \dots, K_{XY} - 1\}$, select normalized features $f^k : \mathcal{X} \rightarrow \mathbb{R}^k$ and $g^k : \mathcal{Y} \rightarrow \mathbb{R}^k$. Given $\delta \geq 0$, assume under each probability measures, $\Phi^{X|U}$ and $\Phi^{Y|V}$ are δ -spherically symmetric. Then,

$$(\bar{E}^{U|S}(f^k), \bar{E}^{V|S}(f^k), \bar{E}^{U|T}(g^k), \bar{E}^{V|T}(g^k)) \leq (C_U \epsilon^2 k, C_V \epsilon^2 \sum_{i=1}^k \sigma_i^2, C_U \epsilon^2 \sum_{i=1}^k \sigma_i^2, C_V \epsilon^2 k) + O(\delta \epsilon^2), \quad (2)$$

as $\epsilon \rightarrow 0$, where C_U and C_V are positive constants independent of ϵ, k or $P_{X,Y}$, and σ_i is the i -th largest singular value of $\tilde{B}_{X,Y}$. Moreover, the equality holds when the feature vectors corresponding to f_i and g_i are precisely the right and left singular vectors of $\tilde{B}_{X,Y}$ corresponding to σ_i .

Remark 2 Specifically, when $\delta = o(1)$ as $\epsilon \rightarrow 0$, (2) reduces to (1), which implies the exact spherical symmetry is unnecessary in Theorem 1.

7. Conclusions

- Theorem 3 recovers Theorem 1, revealing that the exact symmetry condition is unnecessary; rather, weak symmetry suffices, with a small directional preference $\delta = o(1)$ tolerable.
- Even if δ is not adequately small (e.g., $O(1)$), the error bound is well-controlled by and robust against the symmetry deviation δ .
- Our work extends the selection framework to a broad class of approximately symmetric environments. We believe it provides a robust, practical theory for universal feature selection in machine learning and statistical inference.

DETAILS AVAILABLE

D. Tang and G. Han, "Universal Feature Selection with Noisy Observations and Weak Symmetry Conditions," *arXiv preprint arXiv:2605.09396*, 2026.

KEY REFERENCES

- [1] I. Guyon and A. Elisseeff, "An introduction to variable and feature selection," *Journal of machine learning research*, vol. 3, no. Mar, pp. 1157–1182, 2003.
- [2] X. Xu, S.-L. Huang, L. Zheng, and G. W. Wornell, "An information theoretic interpretation to deep neural networks," *Entropy*, vol. 24, no. 1, pp. 135, 2022.
- [3] S.-L. Huang, A. Makur, G. W. Wornell, and L. Zheng, "Universal features for high-dimensional learning and inference," *Foundations and Trends in Communications and Information Theory*, vol. 21, no. 1-2, pp. 1–299, 2024.
- [4] A. P. Dawid, "Spherical matrix distributions and a multivariate model," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 39, no. 2, pp. 254–261, 1977.

MORE INFORMATION



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